

University of Notre Dame

Department of Computer Science and Engineering

CSE 60745

Graph Learning: Techn & Apps

Assignment Date:	
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Student Name:	

HW-2: Part (A) — Solutions

This assignment contains 70 points: Problem 1 (35 pts) and Problem 2 (35 pts).

Problem 1. Benchmarks, baselines and evaluation metrics for graph embedding on *homogeneous* graphs

A *homogeneous* graph $G = (V, E)$ carries a single node type and a single edge type: the type-mapping functions $\phi : V \rightarrow \mathcal{A}$ and $\psi : E \rightarrow \mathcal{R}$ satisfy $|\mathcal{A}| = |\mathcal{R}| = 1$. Optionally each node carries an attribute vector $x_v \in \mathbb{R}^d$ and a label y_v . Graph embedding learns a map $f : V \rightarrow \mathbb{R}^k$ with $k \ll |V|$ such that proximity in $\mathbb{R}^k$ reflects proximity in G ; the embeddings are then evaluated on

- **node classification** — predict y_v for held-out nodes, and
- **link prediction** — score unobserved pairs (u, v) so that true missing edges rank above non-edges.

The heterogeneous counterpart (multiple node/edge types, metapath-based methods) is the subject of Problem 2 and is deliberately not covered here.

Statistics in Table 1 were obtained by loading each graph and counting; the OGB figures in Table 2 are from the official OGB documentation. Where the literature reports conflicting values, the discrepancy is footnoted rather than silently resolved.

1. Benchmark datasets

1.1 Citation networks (the default benchmarks)

Cora, **CiteSeer**, **PubMed** (Sen et al., *Collective Classification in Network Data*, AI Magazine 29(3), 2008; split protocol from Yang, Cohen & Salakhutdinov, *Revisiting Semi-Supervised Learning with Graph Embeddings*, ICML 2016) are the three *Planetoid* citation graphs and remain the default testbed for semi-supervised node classification. Nodes are scientific publications, an undirected edge joins two papers when one cites the other, and the label is the paper’s topic.

Cora collects machine-learning papers and uses **binary bag-of-words** features.¹ **CiteSeer** is the sparsest of the three and has a number of isolated nodes, which is why absolute accuracies are lowest on it. **PubMed** collects diabetes-related biomedical abstracts and, unlike the other two, uses **real-valued TF–IDF** weights rather than binary indicators. All three are also standard *link prediction* benchmarks: the GAE/VGAE protocol hides 5% of edges for validation and 10% for test, samples an equal number of non-edges, and reports AUC and AP.

1.2 Co-authorship and co-purchase networks

Introduced by Shchur et al. (*Pitfalls of Graph Neural Network Evaluation*, NeurIPS 2018 Relational Representation Learning Workshop) precisely because results on the three Planetoid graphs with

¹The Cora edge count is reported inconsistently: 5,429 is the number of lines in the original `cora.cites` file (which contains duplicate and self-referential entries), 5,278 is the number of distinct undirected edges after de-duplication, and 10,556 is the number of *directed* entries stored when each undirected edge is materialised in both directions. All three describe the same graph.

a single fixed split are not statistically reliable.

Coauthor CS and **Coauthor Physics** come from the KDD Cup 2016 Microsoft Academic Graph: an edge means two authors co-authored a paper, features are binary keyword indicators over an author’s papers, and the label is the author’s most active field of study. **Amazon Computers** and **Amazon Photo** are segments of the Amazon co-purchase graph: an edge means two products are frequently bought together, features are binary bag-of-words over product reviews, and the label is the product category. All four are used for node classification and, being attributed and undirected, also serve as link prediction benchmarks.

1.3 Web-page and heterophilous benchmarks (the counterexamples)

The benchmarks above are strongly *homophilous*, and a GNN that simply averages over neighbourhoods does well on them almost by construction. The suite assembled by Pei et al. (*Geom-GCN*, ICLR 2020) and popularised by Zhu et al. (*Beyond Homophily in Graph Neural Networks*, NeurIPS 2020) exists to break that assumption.

WebKB (**Cornell**, **Texas**, **Wisconsin**) are university web pages linked by hyperlinks, with binary bag-of-words features and 5 page classes (*student*, *project*, *course*, *staff*, *faculty*); they are very small and very heterophilous, so variance across splits is large and results must be averaged over the 10 standard splits. **Actor** is the actor-only induced subgraph of a film–director–actor–writer network, where an edge means two actors are co-mentioned on the same Wikipedia page. **Chameleon** and **Squirrel** are Wikipedia page–page networks on those topics, with nodes binned into 5 classes by average monthly traffic. All six are used almost exclusively for node classification.

The quantity that separates the two groups is the **edge homophily ratio**

$$h_{\text{edge}}(G) = \frac{|\{(u, v) \in E : y_u = y_v\}|}{|E|}, \quad (1)$$

which we measured directly on each graph. The values appear in the last column of Table 1 and range from 0.93 (Coauthor Physics) down to 0.11 (Texas); Figure 2 illustrates the two regimes.

The edge set used must be stated, because two choices in it move these numbers.² The ordering of the datasets is unaffected either way: the six web/Wikipedia/co-occurrence graphs sit far below the citation, co-authorship and co-purchase graphs on any of these definitions, which is the only claim the rest of this answer relies on.

²(i) *Which edge set.* We evaluate Eq. (1) over *distinct unordered pairs* — precisely the set counted by the $|E|$ column of Table 1, so the ratio and its denominator refer to the same object. The six directed graphs also admit a count over the stored directed entries, which weights a reciprocated arc twice as heavily as a one-way arc; that gives slightly different values (e.g. Chameleon 0.235 rather than 0.231) and agrees less well with the literature. (ii) *Self-loops.* Texas and Wisconsin carry 16 each, Actor 93 and Squirrel 140, and a self-loop trivially satisfies $y_u = y_v$, so it always counts as homophilous. Following Zhu et al. we include them; excluding them would give Texas 0.06 rather than 0.11 and Wisconsin 0.18 rather than 0.21 — a large relative change on exactly the datasets whose heterophily is the point. Under the stated convention our values reproduce Zhu et al.’s published figures *exactly* to two decimal places for Cora, CiteSeer, PubMed, Texas, Wisconsin, Actor, Chameleon and Squirrel. **Cornell is the sole exception:** Zhu et al. report 0.30 and we measure 0.13. We could not reproduce 0.30 under any definition we tried — over unordered pairs 0.132, the same excluding self-loops 0.123, over the stored directed entries 0.131, or node homophily in the sense of Pei et al. 0.186 — so we report the measured value and flag the disagreement rather than repeating a figure we cannot reproduce.

1.4 Social and interaction networks

These are the benchmarks of the *unsupervised* embedding literature (DeepWalk, LINE, node2vec). They are typically **feature-less** and **multi-label**, which is why Micro-F1/Macro-F1 rather than accuracy became the standard metric in that line of work.

BlogCatalog is a friendship network of bloggers labelled with overlapping interests and **Flickr** is a contact network labelled with the groups each user joined; both are from Tang & Liu, *Relational Learning via Latent Social Dimensions*, KDD 2009. **YouTube** is a user–user social network labelled with group memberships, from the companion paper Tang & Liu, *Scalable Learning of Collective Behavior based on Sparse Social Dimensions*, CIKM 2009, and is the scalability stress test of that literature. (Statistics for all three are as tabulated in the DeepWalk paper.) None has node attributes, so structure is the only signal. **Reddit** (Hamilton, Ying & Leskovec, NIPS 2017) links two posts when the same user commented on both, labels each post with its community, and is the standard *inductive, large-scale* node-classification benchmark.³

Warning — three names, six graphs. *BlogCatalog*, *Flickr* and *PPI* each denote two genuinely different graphs in this literature, and numbers are frequently mixed up between them. The feature-less multi-label versions above are the ones used by DeepWalk/node2vec; the attributed versions (BlogCatalog with 5,196 nodes / 171,743 edges / 8,189 features / 6 classes; the GraphSAINT Flickr with 89,250 nodes / 449,878 edges / 500 features / 7 classes) are the ones shipped by modern GNN libraries. A paper’s numbers are only comparable to another’s if the same version is used.

1.5 Biological networks

PPI (protein–protein interaction, GraphSAGE version) is **24 separate graphs**, one per human tissue, whose node features are positional gene sets, motif gene sets and immunological signatures, and whose **121 binary labels** are gene ontology terms. The split is *inductive across graphs* — 20 train, 2 validate, 2 test — so the test graphs are never seen during training. This is the canonical test of whether a method generalises to unseen graphs at all, which transductive methods such as DeepWalk cannot do by construction. The **node2vec PPI** is a different graph: a single *Homo sapiens* subgraph with 3,890 nodes, 76,584 edges and 50 labels.

1.6 The Open Graph Benchmark (OGB)

Hu et al. (*Open Graph Benchmark: Datasets for Machine Learning on Graphs*, NeurIPS 2020) introduced OGB to fix the two standing problems with the classic benchmarks: they are tiny, and their random splits are unrealistically easy. OGB fixes the split, fixes the metric, and provides a shared evaluator, so numbers are comparable across papers. Statistics are in Table 2. The homogeneous members are:

- **ogbn-arxiv** — directed citation graph of CS arXiv papers, node features the averaged word2vec embedding of title and abstract, labels the 40 subject areas. *Time split*: train up to 2017, validate on 2018, test from 2019.
- **ogbn-products** — Amazon co-purchase graph, bag-of-words-plus-PCA features, 47 top-level categories. *Sales-rank split*: the most popular products train and the rest are validation/test,

³We measured 57,307,946 distinct undirected edges (114,615,892 directed entries). This reproduces the average degree of 492 stated in the GraphSAGE paper ($2 \times 57,307,946 / 232,965 = 492.0$); the figure of 11.6M edges that appears on some dataset pages refers to a sparsified copy of the graph.

which is much harder than a random split.

- **ogbn-papers100M** — the same construction as ogbn-arxiv at 111M nodes; only ~ 1.5 M are labelled (172 subject areas). The standard scalability benchmark.
- **ogbn-proteins** — proteins from 8 species joined by weighted association edges. It carries *edge* features and *no* node features, and its 112 binary function-prediction tasks are split *by species* and scored by average ROC-AUC.
- **ogbl-collab** (author collaboration), **ogbl-ppa** (protein association), **ogbl-citation2** (paper citation) and **ogbl-ddi** (drug–drug interaction) are the four homogeneous *link prediction* tasks. Each fixes its own split, metric and negative set in Table 2; ogbl-ddi is notable for having *no* node features at all, so only structure can be used.

1.7 Split protocols — and why they must be stated

A result is meaningless without its split, because the same model can differ by many accuracy points across protocols on the same graph:

- **Planetoid semi-supervised split:** exactly **20 labelled nodes per class** for training, 500 for validation, 1,000 for test. This is the split on which the familiar Cora $\approx 81.5\%$ figure for GCN is reported.
- **Full-supervised / “Geom-GCN” split:** **60/20/20%** of nodes per class, averaged over 10 random splits. Standard for the heterophilous benchmarks and typically 5–10 points higher than the 20-per-class numbers.
- **Random splits with fixed label budget** (Shchur et al.): the Planetoid budget but re-randomised, reported as mean $\pm$ std over ≥ 100 splits. This was the paper’s central recommendation, since the single fixed Planetoid split does not support the ranking of methods usually claimed from it.
- **OGB splits** are fixed and non-random — by time, by sales rank, by species, by experimental throughput, by protein target — and deliberately induce distribution shift between train and test.
- **Link prediction:** edges, not nodes, are split. The message-passing graph must contain *only* training edges, otherwise the held-out edge leaks into its own prediction. See Figure 3.

2. Graph schemas of the benchmarks

The *graph schema* is the meta-template of a network: the graph over node types whose edges are the permitted relation types. For a homogeneous graph the schema is by definition **degenerate** — one node type and one edge type — so it is a single vertex with a single self-loop:

$$\mathcal{T}_G = (\{A\}, \{A \xrightarrow{R} A\}), \quad |\mathcal{A}| = |\mathcal{R}| = 1.$$

This is the honest answer to “draw the related graph schema” for these benchmarks, and it is exactly what distinguishes them from the heterogeneous datasets of Problem 2, whose schemas have several node types and several relation types. What actually differs between the homogeneous benchmarks is the *semantics* instantiated into A and R , so the benchmarks are grouped into seven schema families in Figure 1. Because a one-node schema is visually vacuous on its own, Figure 2 additionally sketches the local structure that the schema does *not* capture and which nevertheless drives model performance: the label homophily of Eq. (1).

Table 1: Homogeneous node-classification / link-prediction benchmarks. **dir.:** **U** = the stored edge set is symmetric (undirected), **D** = it is not (directed). $|E|$ counts *distinct unordered pairs*, with self-loops counted once; the parenthesised figure is the number of entries in the stored edge list, which is $2|E|$ for the U rows but not for the D rows, since a directed graph stores only the arcs that exist. h_{edge} is Eq. (1) measured on the graph, its denominator being exactly the $|E|$ of the preceding column — the same distinct unordered pairs, self-loops included (see the footnote; this convention matters, and the alternatives are listed there). “bin.” = binary bag-of-words, “TF-IDF”/“real” = real-valued, ML = multi-label. [†]Cornell: see footnote 2.

Family	Dataset	$ V $	$ E $	d	#cls	feat.	dir.	h_{edge}
Citation	Cora	2,708	5,278 (10,556)	1,433	7	bin.	U	0.81
	CiteSeer	3,327	4,552 (9,104)	3,703	6	bin.	U	0.74
	PubMed	19,717	44,324 (88,648)	500	3	TF-IDF	U	0.80
Co-author	Coauthor CS	18,333	81,894	6,805	15	bin.	U	0.81
	Coauthor Physics	34,493	247,962	8,415	5	bin.	U	0.93
Co-purchase	Amazon Computers	13,752	245,861	767	10	bin.	U	0.78
	Amazon Photo	7,650	119,081	745	8	bin.	U	0.83
Web / hyperlink	Cornell	183	280 (298)	1,703	5	bin.	D	0.13 [†]
	Texas	183	295 (325)	1,703	5	bin.	D	0.11
	Wisconsin	251	466 (515)	1,703	5	bin.	D	0.21
	Chameleon	2,277	31,421 (36,101)	2,325	5	bin.	D	0.23
	Squirrel	5,201	198,493 (217,073)	2,089	5	bin.	D	0.22
Co-occurrence	Actor	7,600	26,752 (30,019)	932	5	bin.	D	0.22
Social	BlogCatalog	10,312	333,983	—	39	none (ML)	U	—
	Flickr	80,513	5,899,882	—	195	none (ML)	U	—
	YouTube	1,138,499	2,990,443	—	47	none (ML)	U	—
	Reddit	232,965	57,307,946	602	41	real	U	—
Biological	PPI (24 graphs)	56,944	793,632	50	121	real (ML)	U	—

Table 2: The homogeneous Open Graph Benchmark datasets. Edge counts are as published by OGB (undirected counts for undirected graphs).

Dataset	Domain	$ V $	$ E $	Node features	Split	Metric
<i>Node classification</i>						
ogbn-arxiv	citation	169,343	1,166,243	128 word2vec	time	Accuracy
ogbn-products	co-purchase	2,449,029	61,859,140	100 BoW+PCA	sales rank	Accuracy
ogbn-papers100M	citation	111,059,956	1,615,685,872	128 word2vec	time	Accuracy
ogbn-proteins	protein	132,534	39,561,252	none (8-d edges)	species	ROC-AUC
<i>Link prediction</i>						
ogbl-collab	co-authorship	235,868	1,285,465	128 word2vec	time	Hits@50
ogbl-ppa	protein	576,289	30,326,273	58-d one-hot	throughput	Hits@100
ogbl-citation2	citation	2,927,963	30,561,187	128 word2vec	time	MRR
ogbl-ddi	drug-drug	4,267	1,334,889	none	protein target	Hits@20

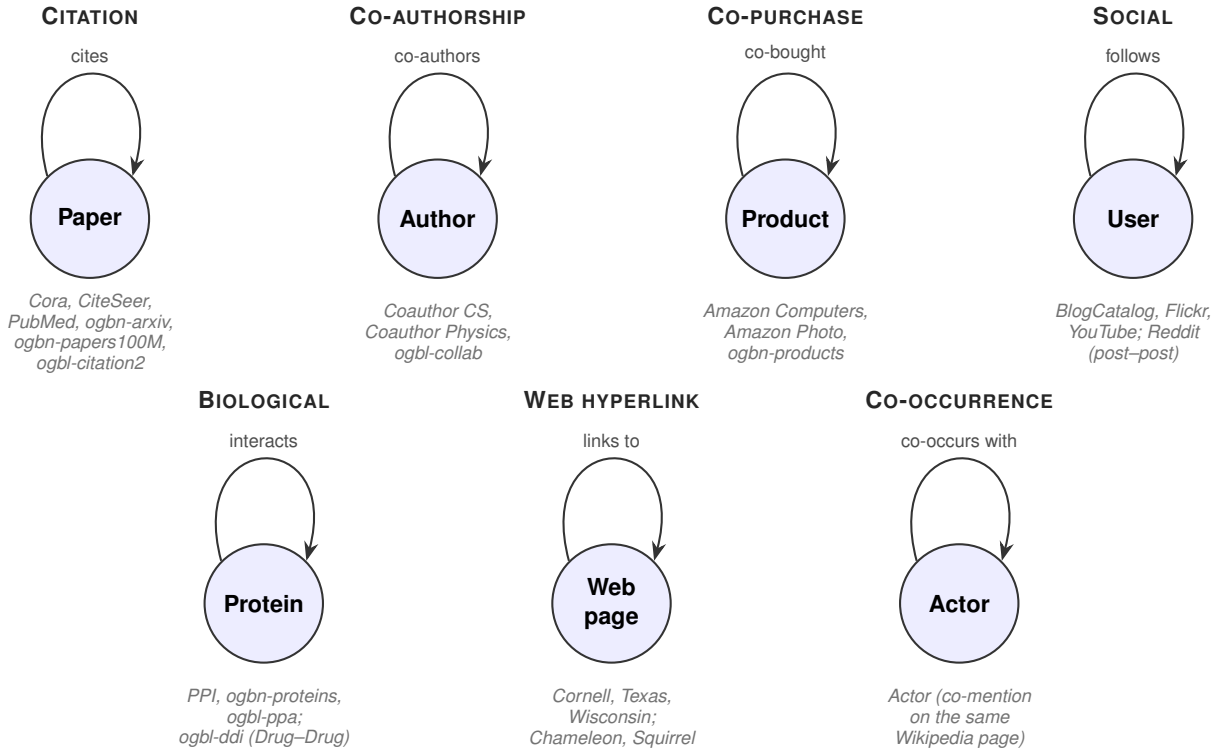


Figure 1: Graph schemas of the homogeneous benchmarks, grouped into seven families. Every schema is a single node type with a single self-relation; only the interpretation of the node and the edge changes between families.

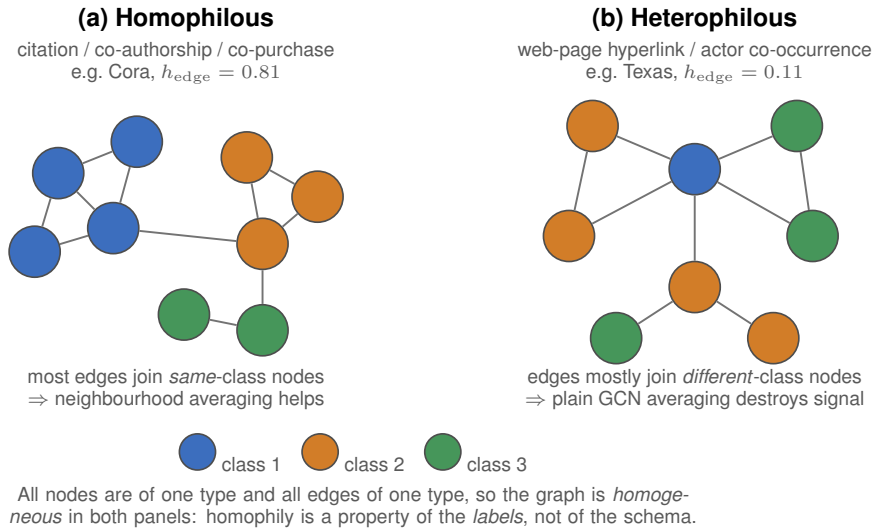


Figure 2: Illustrative subgraphs. Both graphs have the same (degenerate) schema; they differ in how labels are distributed over edges. The measured h_{edge} values are those of Table 1.

3. Baselines

3.1 Matrix factorisation and spectral methods

- **Laplacian Eigenmaps** — Belkin & Niyogi, *Laplacian Eigenmaps and Spectral Techniques for Embedding and Clustering*, NIPS 2001; expanded as *Laplacian Eigenmaps for Dimensionality Reduction and Data Representation*, *Neural Computation* 15(6):1373–1396, 2003 (the canonical citation). Embeds nodes by the eigenvectors of the graph Laplacian $L = D - A$ belonging to the smallest non-zero eigenvalues, i.e. it minimises $\sum_{(u,v) \in E} w_{uv} \|f(u) - f(v)\|^2$ subject to an orthogonality constraint ruling out the trivial solution. The ancestor of essentially every proximity-preserving embedding; spectral clustering is the same computation.
- **GraRep** — Cao, Lu & Xu, *GraRep: Learning Graph Representations with Global Structural Information*, CIKM 2015. Observes that first-order proximity is not enough, and factorises the k -step transition matrices $P^1, \dots, P^K$ (via SVD of their log-transformed positive entries) separately, concatenating the K resulting embeddings. This captures k -hop structure explicitly, at the cost of dense $|V| \times |V|$ powers.
- **HOPE** — Ou, Cui, Pei, Zhang & Zhu, *Asymmetric Transitivity Preserving Graph Embedding*, KDD 2016. Targets *directed* graphs, where proximity is asymmetric. It learns two vectors per node (source and target) and factorises a high-order proximity matrix — Katz, Rooted PageRank, Common Neighbors or Adamic–Adar — which it shows can be written in a sparse form $S = M_g^{-1} M_\ell$ admitting a generalised SVD, so the cost is linear in $|E|$.
- **NetMF** — Qiu, Dong, Ma, Li, Wang & Tang, *Network Embedding as Matrix Factorization: Unifying DeepWalk, LINE, PTE, and node2vec*, WSDM 2018. Proves that the skip-gram-with-negative-sampling objectives of DeepWalk, LINE, PTE and node2vec all implicitly factorise a closed-form matrix, for DeepWalk $\log(\frac{\text{vol}(G)}{bT} (\sum_{r=1}^T P^r) D^{-1})$. Factorising that matrix directly gives better embeddings than running the sampler, and the paper is the standard theoretical bridge between the random-walk and factorisation families.

3.2 Random-walk / skip-gram methods

- **DeepWalk** — Perozzi, Al-Rfou & Skiena, *DeepWalk: Online Learning of Social Representations*, KDD 2014. The paper that started the field. It samples uniform random walks from each node, treats each walk as a “sentence” and each node as a “word”, and runs word2vec/skip-gram with hierarchical softmax over them. Nodes appearing in similar walk contexts receive similar vectors. It requires no node attributes and is trivially parallel, but is transductive: unseen nodes have no embedding.
- **LINE** — Tang, Qu, Wang, Zhang, Yan & Mei, *LINE: Large-scale Information Network Embedding*, WWW 2015. Dispenses with walks and optimises two explicit objectives: *first-order* proximity (observed edges, $p_1(u, v) = \sigma(f(u)^\top f(v))$) and *second-order* proximity (shared neighbourhoods, a context embedding per node), each trained by negative sampling with edge-sampling to handle weights. The two embeddings are concatenated. It scales to millions of nodes and is the standard “no-walk” baseline.
- **node2vec** — Grover & Leskovec, *node2vec: Scalable Feature Learning for Networks*, KDD 2016. Generalises DeepWalk by biasing the walk with return parameter p and in-out parameter q ,

interpolating between breadth-first search (which captures *structural equivalence*) and depth-first search (which captures *homophily*). Tuning (p, q) per dataset is what buys its consistent gain over DeepWalk. It also defines the standard Hadamard/L1/L2 binary operators for turning node embeddings into edge features for link prediction.

- **struc2vec** — Ribeiro, Savarese & Figueiredo, *struc2vec: Learning Node Representations from Structural Identity*, KDD 2017. Argues that nodes with similar *roles* should embed together even if they are far apart in the graph — something no proximity-based method can do. It builds a multilayer context graph whose layer- k edge weights compare the ordered degree sequences of k -hop neighbourhoods, then runs skip-gram on walks over that auxiliary graph.

3.3 Autoencoder methods

- **SDNE** — Wang, Cui & Zhu, *Structural Deep Network Embedding*, KDD 2016. A deep autoencoder whose input is a node's adjacency row. It combines a reconstruction loss on that row (second-order proximity), re-weighted so that the many zeros do not dominate the few ones, with a Laplacian-eigenmaps-style penalty tying connected nodes together (first-order proximity). The first non-linear (deep) network embedding method.
- **GAE / VGAE** — Kipf & Welling, *Variational Graph Auto-Encoders*, NIPS 2016 Bayesian Deep Learning Workshop. Uses a two-layer GCN as the encoder, $Z = \text{GCN}(X, A)$, and a simple inner-product decoder $\hat{A} = \sigma(ZZ^\top)$, trained by reconstruction cross-entropy; the variational version places a Gaussian prior on Z and adds a KL term. It is the default *unsupervised* baseline for link prediction on Cora/CiteSeer/PubMed.

3.4 Graph neural networks (message passing)

All of these compute $h_v^{(l+1)} = \text{UPDATE}(h_v^{(l)}, \text{AGGREGATE}(\{h_u^{(l)} : u \in \mathcal{N}(v)\}))$ and differ in the two functions.

- **GCN** — Kipf & Welling, *Semi-Supervised Classification with Graph Convolutional Networks*, ICLR 2017. A first-order approximation of a Chebyshev spectral filter, giving the layer $H^{(l+1)} = \sigma(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(l)} W^{(l)})$ with $\tilde{A} = A + I$. It is a symmetrically-normalised neighbourhood average followed by a linear map; two layers is the standard configuration. The single most reported baseline in the field.
- **GraphSAGE** — Hamilton, Ying & Leskovec, *Inductive Representation Learning on Large Graphs*, NIPS 2017. Makes GNNs *inductive* and scalable: instead of using the full adjacency it samples a fixed number of neighbours per hop and learns an aggregator (mean, LSTM or max-pool) plus a concatenation with the node's own previous representation. Because the aggregator is shared, embeddings can be produced for nodes — and graphs — never seen in training. Introduced the Reddit and PPI benchmarks.
- **GAT** — Veličković, Cucurull, Casanova, Romero, Liò & Bengio, *Graph Attention Networks*, ICLR 2018. Replaces the fixed normalisation coefficient $1/\sqrt{d_u d_v}$ with a learned attention weight $\alpha_{uv} = \text{softmax}_u(\text{LeakyReLU}(a^\top [Wh_v || Wh_u]))$, so a node can weight its neighbours unequally. Multi-head attention stabilises training. No spectral decomposition and no knowledge of the full graph is required, so it too is inductive.

- **JK-Net** — Xu, Li, Tian, Sonobe, Kawarabayashi & Jegelka, *Representation Learning on Graphs with Jumping Knowledge Networks*, ICML 2018. Observes that the right neighbourhood radius differs per node (hub vs. periphery), and lets the final representation “jump” by combining *all* intermediate layers $h^{(1)}, \dots, h^{(L)}$ via concatenation, max-pooling or an LSTM attention, rather than using only the last. A standard remedy for over-smoothing.
- **GIN** — Xu, Hu, Leskovec & Jegelka, *How Powerful are Graph Neural Networks?*, ICLR 2019. Analyses message passing through the Weisfeiler–Lehman test and shows that mean and max aggregation lose information; a *sum* aggregator with an MLP update, $h_v^{(l+1)} = \text{MLP}((1 + \epsilon)h_v^{(l)} + \sum_{u \in \mathcal{N}(v)} h_u^{(l)})$, is maximally expressive, i.e. as strong as 1-WL. The reference point for all expressive-power arguments.
- **SGC** — Wu, Souza, Zhang, Fifty, Yu & Weinberger, *Simplifying Graph Convolutional Networks*, ICML 2019. Removes every non-linearity from a K -layer GCN, collapsing it to $\hat{Y} = \text{softmax}(S^K XW)$ with $S = \tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2}$. Since $S^K X$ is a fixed pre-computation, training is plain logistic regression — orders of magnitude faster, and on the homophilous citation benchmarks essentially as accurate. An important sanity baseline: it measures how much the non-linearity was really contributing.
- **APPNP** — Gasteiger (formerly Klicpera), Bojchevski & Günnemann, *Predict then Propagate: Graph Neural Networks meet Personalized PageRank*, ICLR 2019. Decouples prediction from propagation: an MLP produces per-node predictions H , which are then diffused by a truncated personalized-PageRank iteration $Z^{(k+1)} = (1 - \alpha)\hat{A}Z^{(k)} + \alpha H$. The teleport term α keeps the node’s own signal alive, so a large propagation radius is possible without over-smoothing and without extra parameters.
- **GCNII** — Chen, Wei, Huang, Ding & Li, *Simple and Deep Graph Convolutional Networks*, ICML 2020. Makes genuinely deep GCNs (32–64 layers) work using two ingredients: an *initial residual* connection to $H^{(0)}$ (the APPNP teleport idea) and *identity mapping* of the weight matrix, $(1 - \beta_l)I + \beta_l W^{(l)}$. Proves the result still expresses a polynomial filter of arbitrary order and so does not collapse to over-smoothing.
- **SEAL** — Zhang & Chen, *Link Prediction Based on Graph Neural Networks*, NeurIPS 2018. Reframes link prediction as *subgraph classification*: for a candidate pair (u, v) it extracts the h -hop enclosing subgraph, applies a Double-Radius Node Labeling that encodes each node’s distance to both u and v , and classifies the labelled subgraph with a GNN. The γ -decaying theory in the paper shows such a GNN can learn any of the classical heuristics below rather than assuming one. The strongest general-purpose link-prediction baseline.
- **NBFNet** — Zhu, Zhang, Xhonneux & Tang, *Neural Bellman-Ford Networks: A General Graph Neural Network Framework for Link Prediction*, NeurIPS 2021. Generalises the Bellman–Ford path formulation by replacing $(\min, +)$ with learned INDICATOR, MESSAGE and AGGREGATE operators, so the score for (u, v) is a learned sum over all paths from u to v . Inductive (it depends on structure relative to u , not on node identity) and applies to homogeneous graphs and knowledge graphs alike.

3.5 Self-supervised and contrastive methods

- **DGI** — Veličković, Fedus, Hamilton, Liò, Bengio & Hjelm, *Deep Graph Infomax*, ICLR 2019. Maximises the mutual information between patch representations h_v and a global summary s

of the graph, using a discriminator trained to tell real (h_v, s) pairs from corrupted ones (the corruption shuffles the node features). Needs no random walks and no labels, and was the first unsupervised method to match supervised GCN on the citation benchmarks.

- **GRACE** — Zhu, Xu, Yu, Liu, Wu & Wang, *Deep Graph Contrastive Representation Learning*, ICML 2020 Workshop on Graph Representation Learning and Beyond. A node-level SimCLR analogue: generate two views by removing edges and masking features, then maximise agreement between the two representations of the same node against all other nodes as negatives (InfoNCE). Contrasts at node rather than graph-summary level, which DGI does not.
- **GraphCL** — You, Chen, Sui, Chen, Wang & Shen, *Graph Contrastive Learning with Augmentations*, NeurIPS 2020. Systematically studies four graph augmentations — node dropping, edge perturbation, attribute masking and subgraph sampling — and shows which combinations help on which domain. Mostly graph-level, but its augmentation study is the reference for the whole contrastive family.
- **BGRL** — Thakoor, Tallec, Azar, Azabou, Dyer, Munos, Veličković & Valko, *Large-Scale Representation Learning on Graphs via Bootstrapping*, ICLR 2022. Removes negative samples altogether using a BYOL-style online/target encoder pair with a stop-gradient, cutting memory from quadratic to linear in the number of nodes. This is what makes self-supervised learning feasible at OGB scale.

3.6 Non-learned link-prediction heuristics

Every link-prediction paper reports these; they are strong, free, and a learned method that does not beat them is not worth its compute. The framing is due to Liben-Nowell & Kleinberg (*The Link Prediction Problem for Social Networks*, CIKM 2003; journal version JASIST 58(7), 2007). Let $\Gamma(u)$ be the neighbour set of u and A the adjacency matrix.

Heuristic	Score $s(u, v)$	Reference / note
Common Neighbors	$ \Gamma(u) \cap \Gamma(v) $	Liben-Nowell & Kleinberg, CIKM 2003
Jaccard	$ \Gamma(u) \cap \Gamma(v) / \Gamma(u) \cup \Gamma(v) $	normalises CN by degree
Adamic-Adar	$\sum_{w \in \Gamma(u) \cap \Gamma(v)} 1 / \log \Gamma(w) $	Adamic & Adar, <i>Friends and Neighbors on the Web</i> , Social Networks 25(3), 2003
Resource Allocation	$\sum_{w \in \Gamma(u) \cap \Gamma(v)} 1 / \Gamma(w) $	Zhou, Lü & Zhang, <i>Eur. Phys. J. B</i> 71(4), 2009; penalises hubs harder than AA
Preferential Attachment	$ \Gamma(u) \cdot \Gamma(v) $	Barabási & Albert, <i>Science</i> 286, 1999
Katz	$\sum_{\ell \geq 1} \beta^\ell \text{paths}_{u,v}^{(\ell)} = ((I - \beta A)^{-1} - I)_{uv}$	Katz, <i>Psychometrika</i> 18(1), 1953; global, needs $\beta < 1/\lambda_{\max}$
Personalized PageRank	stationary probability at v of a random walk from u with restart	Jeh & Widom, <i>Scaling Personalized Web Search</i> , WWW 2003; PageRank itself: Page, Brin, Motwani & Winograd, Stanford InfoLab TR, 1999
SimRank	$\frac{\gamma}{ \Gamma(u) \Gamma(v) } \sum_{a \in \Gamma(u)} \sum_{b \in \Gamma(v)} s(a, b)$	Jeh & Widom, KDD 2002; “similar if referenced by similar nodes”

The first five are *local* (2-hop) and therefore cheap but blind to long-range structure; Katz, PPR and SimRank are *global* and stronger but cost a matrix inverse or an iterative solve. On ogbl-ddi and ogbl-ppa, Common Neighbors and Adamic–Adar remain competitive with many GNNs, which is the usual argument for reporting them.

4. Evaluation metrics

4.1 Node classification

Let C be the number of classes and TP_c, FP_c, FN_c the per-class counts.

- **Accuracy** = $\frac{1}{|V_{\text{test}}|} \sum_v \mathbf{1}[\hat{y}_v = y_v]$. The default for single-label benchmarks (Planetoid, OGB node tasks). Misleading under class imbalance.
- **Precision / Recall / F1** per class: $P_c = \frac{TP_c}{TP_c + FP_c}$, $R_c = \frac{TP_c}{TP_c + FN_c}$, $F1_c = \frac{2P_c R_c}{P_c + R_c}$.
- **Micro-F1** — pool the confusion matrix over all classes first, then compute one F1:

$$\text{Micro-F1} = \frac{2 \sum_c TP_c}{2 \sum_c TP_c + \sum_c FP_c + \sum_c FN_c}. \quad (2)$$

In single-label multi-class classification every error is simultaneously one FP and one FN, so **Micro-F1 = Accuracy**. It is dominated by the frequent classes.

- **Macro-F1** — the *unweighted* mean of the per-class F1 scores:

$$\text{Macro-F1} = \frac{1}{C} \sum_{c=1}^C F1_c. \quad (3)$$

Because every class contributes equally regardless of size, a model that ignores a rare class is punished. **This is why Macro-F1 matters**: on BlogCatalog or Flickr, where label frequencies span orders of magnitude, a model can score well on Micro-F1 while never predicting most of the 39 (resp. 195) labels — the gap between the two numbers is the diagnostic, and reporting only Micro-F1 hides it.

- **ROC-AUC**, computed one-vs-rest and averaged over classes for multi-class problems, and averaged over tasks for multi-label problems (this is the ogbn-proteins metric). It is threshold-free, so it measures ranking quality rather than the quality of a particular decision threshold.
- **Reporting convention**. A single number is not a result. The convention is **mean $\pm$ standard deviation over 10 (OGB, fixed split, 10 seeds) or 100 (Shchur et al., random splits) runs**, since the spread across splits on the small benchmarks routinely exceeds the differences between published methods.
- **Protocol for unsupervised embeddings**. DeepWalk/LINE/node2vec/DGI are evaluated by the *linear-probe* protocol: learn the embedding with no labels, **freeze** it, train a one-vs-rest ℓ_2 -regularised logistic regression on the embeddings of a fraction ρ of nodes, and report Micro-F1 and Macro-F1 on the rest. Results are given as a curve over training ratios $\rho \in \{10\%, \dots, 90\%\}$ (BlogCatalog) or $\{1\%, \dots, 10\%\}$ (Flickr, YouTube). Freezing is essential — fine-tuning the encoder would measure something else.

4.2 Link prediction

Link prediction is extremely class-imbalanced (a graph with $|V|$ nodes has $\binom{|V|}{2} - |E|$ non-edges), so the metric choice matters far more than for node classification. Positives are held-out edges; negatives are sampled node pairs that are not edges (Figure 3).

- **AUC-ROC** — the probability that a random positive outranks a random negative. Standard in the GAE/VGAE protocol with a 1:1 positive:negative ratio.
- **Average Precision (AP / AUPRC)** —

$$\text{AP} = \sum_k (R_k - R_{k-1}) P_k, \quad (4)$$

with P_k, R_k the precision and recall at threshold rank k . Unlike AUC it does not credit a model for correctly ranking the vast easy majority of negatives.

- **Precision@K / Recall@K** — the fraction of the top- K predicted pairs that are true edges, and the fraction of true edges recovered in the top K .
- **Hits@K** — the fraction of positive edges ranked in the top K against the sampled negatives:

$$\text{Hits@K} = \frac{1}{|E_{\text{test}}|} \sum_{e \in E_{\text{test}}} \mathbf{1}[\text{rank}(e) \leq K]. \quad (5)$$

OGB fixes K per dataset so that numbers are comparable: **Hits@50** for ogbl-collab (against 100,000 shared negatives), **Hits@20** for ogbl-ddi, **Hits@100** for ogbl-ppa (against 3,000,000 negatives).

- **MRR** — for each query (in ogbl-citation2, each source paper, ranked against 1,000 sampled negative references),

$$\text{MRR} = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{\text{rank}_q}, \quad (6)$$

where rank_q is the position of the first true item for query q . Rewards getting the answer at the very top far more than at position 10.

- **NDCG@K** — with rel_i the relevance of the item at rank i ,

$$\text{DCG@K} = \sum_{i=1}^K \frac{2^{\text{rel}_i} - 1}{\log_2(i + 1)}, \quad \text{NDCG@K} = \frac{\text{DCG@K}}{\text{IDCG@K}}, \quad (7)$$

IDCG@K being the DCG of the ideal ordering. Used when a node has many true links of differing importance (recommendation-style evaluation).

Negative sampling, and why AUC is optimistic. The negatives define the metric. Three conventions are in use: (i) *1:1 random negatives* (the GAE protocol), (ii) *a shared fixed negative set* for all positives (ogbl-collab, ogbl-ddi, ogbl-ppa), and (iii) *per-query negatives* (ogbl-citation2: 1,000 previously-published papers per source node). Real graphs are sparse, so a uniformly sampled pair is almost always a pair of distant, low-degree nodes that any model rejects trivially. AUC averages over all such pairs and therefore saturates near 1.0 and stops discriminating between methods —

this is precisely why OGB abandoned it in favour of Hits@K and MRR, which only reward ranking the positive above the *hardest* sampled negatives. AP is likewise more informative than AUC because it weights the high-precision region where the model actually operates. A corollary is that Hits@K numbers are *only* comparable when the negative set is identical, which is the reason OGB ships a fixed evaluator.

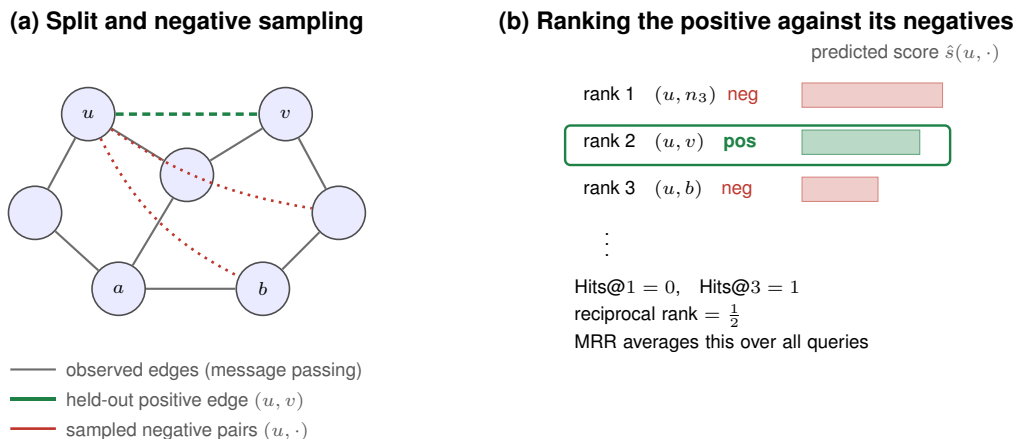


Figure 3: The link-prediction evaluation protocol. The held-out positive edge must be removed from the message-passing graph, and is scored against a sampled set of negatives; Hits@K and MRR are read off the resulting ranking.

4.3 Which metric is reported on which dataset

5. Summary

For homogeneous graphs the schema is degenerate — one node type, one edge type (Figure 1) — so what distinguishes the benchmarks is scale, feature type, and above all label homophily (Eq. (1)). The suite has moved in three waves: the small homophilous citation graphs with a fixed 20-per-class split, the re-splittable co-authorship/co-purchase graphs and the heterophilous WebKB/Wikipedia graphs introduced to expose the homophily assumption, and finally OGB, with realistic non-random splits and a shared evaluator at a scale where sampling-based training is mandatory. On metrics, the two points that matter most are that Micro-F1 equals accuracy in the single-label case while Macro-F1 is what exposes neglected rare classes, and that on sparse graphs AUC is optimistic, so ranking metrics against a *fixed* negative set (Hits@K, MRR) are what make link-prediction results comparable.

Table 3: Dataset $\rightarrow$ task $\rightarrow$ metric normally reported.

Dataset	Task	Metric normally reported
Cora / CiteSeer / PubMed	node classification	Accuracy (= Micro-F1), mean $\pm$ std over 10–100 splits
Cora / CiteSeer / PubMed	link prediction	AUC-ROC and AP (GAE protocol, 85/5/10 edge split)
Coauthor CS / Physics	node classification	Accuracy, mean $\pm$ std over 100 random splits
Amazon Computers / Photo	node classification	Accuracy, mean $\pm$ std over 100 random splits
WebKB, Actor, Chameleon, Squirrel	node classification	Accuracy over the 10 fixed 60/20/20 splits
BlogCatalog / Flickr / YouTube	node classification, multi-label	Micro-F1 and Macro-F1 vs. training ratio
PPI (24 graphs)	inductive node classification, multi-label	Micro-F1
Reddit	inductive node classification	Micro-F1 (= Accuracy)
ogbn-arxiv / products / papers100M	node classification	Accuracy (fixed OGB split)
ogbn-proteins	node classification, 112 tasks	ROC-AUC averaged over tasks
ogbl-collab	link prediction	Hits@50 (100,000 shared negatives)
ogbl-ddi	link prediction	Hits@20 ($\sim$ 100,000 shared negatives)
ogbl-ppa	link prediction	Hits@100 (3,000,000 shared negatives)
ogbl-citation2	link prediction	MRR (1,000 negatives per source node)

Problem 2: Benchmarks, Baselines and Evaluation Metrics for Heterogeneous Graph Embedding

This answer covers graph embedding on *heterogeneous* graph structures for node classification (NC) and link prediction (LP). The homogeneous case is treated in Problem 1; homogeneous methods appear here only in their role as type-agnostic baselines.

1. Set-up: HINs, network schemas and meta-paths

Heterogeneous information network (HIN). Following Sun et al. (*PathSim*, PVLDB 4(11):992–1003, 2011) and the survey of Shi, Li, Zhang, Sun & Yu (*A Survey of Heterogeneous Information Network Analysis*, IEEE TKDE 29(1):17–37, 2017), a **heterogeneous information network** is a graph

$$G = (V, E, \phi, \psi), \quad \phi : V \rightarrow \mathcal{A}, \quad \psi : E \rightarrow \mathcal{R},$$

in which every node $v \in V$ carries exactly one *node type* $\phi(v)$ from the type set $\mathcal{A}$, every edge $e \in E$ carries exactly one *relation type* $\psi(e)$ from $\mathcal{R}$, and

$$|\mathcal{A}| + |\mathcal{R}| > 2$$

This inequality *is* the definition of heterogeneity: when $|\mathcal{A}| = |\mathcal{R}| = 1$ the network degenerates to the homogeneous graph of Problem 1. Each relation is typed at both ends, inducing a map $A_i \xrightarrow{R} A_j$ between two specific node types; its inverse R^{-1} is in general a *different* relation (*writes* vs. *written-by*), which is why the released benchmarks below count 6 or 8 “edge types” where a reader might expect 3 or 4.

Network schema. The **network schema** $T_G = (\mathcal{A}, \mathcal{R})$ is the directed graph whose vertices are the node types $\mathcal{A}$ and whose edges are the relation types $\mathcal{R}$. It is a meta-template of which G is an instantiation: every node and edge of G maps to some type in T_G , so the schema states which kinds of connection the dataset can express at all. This is the object Section 3 draws for each benchmark. Note that the drawing carries real information here whereas it does not in the homogeneous case: Cora’s schema is a single node type with one self-relation, while DBLP’s is a four-type star.

Meta-path. A **meta-path** $\mathcal{P}$ of length ℓ is a path defined *on the schema*,

$$\mathcal{P} : A_1 \xrightarrow{R_1} A_2 \xrightarrow{R_2} \dots \xrightarrow{R_\ell} A_{\ell+1},$$

denoting the composite relation $R = R_1 \circ R_2 \circ \dots \circ R_\ell$ from type A_1 to type $A_{\ell+1}$. A node sequence $v_1 v_2 \dots v_{\ell+1}$ in G with $\phi(v_i) = A_i$ and $\psi(v_i v_{i+1}) = R_i$ is a *path instance*. Meta-paths are written by type initials, so on DBLP APA abbreviates Author $\xrightarrow{\text{writes}}$ Paper $\xrightarrow{\text{written-by}}$ Author.

Meta-paths matter because each defines a *different* homogeneous projection of the same HIN, and the projections carry different semantics: on DBLP APA means *co-authorship* while APCPA means *published at the same venue*, and two authors may be strongly related under one and unrelated under the other. Nearly every method in Section 4 is a different answer to “how should meta-path structure be used?” — fixed and hand-chosen (metapath2vec, HAN), aggregated with attention (MAGNN), learned (GTN), or discarded in favour of per-relation parameters (R-GCN, HGT, simple-HGN).

2. Benchmarks

2.1 The standardised suite: HGB

Heterogeneous GNN papers before 2021 each preprocessed DBLP/ACM/IMDB their own way, so their reported numbers were not comparable. **HGB** (Heterogeneous Graph Benchmark; Lv, Ding, Liu et al., KDD 2021) exists precisely to fix this: fixed splits, fixed features, one leaderboard. Its headline statistics are Table 4; the per-type breakdown is Table 5.

Table 4: HGB (Lv et al., KDD 2021) benchmark statistics, as published.

Task	Dataset	#Nodes	#N. types	#Edges	#E. types	Target	#Classes
NC	DBLP	26,128	4	239,566	6	author	4
	IMDB	21,420	4	86,642	6	movie	5 [†]
	ACM	10,942	4	547,872	8	paper	3
	Freebase	180,098	8	1,057,688	36	book	7
LP	Amazon	10,099	1	148,659	2	product-product	—
	LastFM	20,612	3	141,521	3	user-artist	—
	PubMed	63,109	4	244,986	10	disease-disease	—

[†] **multi-label** — see Section 5.

Table 5: Per-node-type and per-relation breakdown of the four HGB node-classification benchmarks, recomputed from the released data files (see the note below).

Dataset	Node type	Count	Feat. dim	Relation	#Edges
DBLP	author *	4,057	334	author-paper	19,645
	paper	14,328	4,231	paper-term	85,810
	term	7,723	50	paper-venue	14,328
	venue	20	—	(+3 inverse relations)	
IMDB	movie *	4,932	3,489	movie-director	4,932
	director	2,393	3,341	movie-actor	14,779
	actor	6,124	3,341	movie-keyword	23,610
	keyword	7,971	—	(+3 inverse relations)	
ACM	paper *	3,025	1,902	paper-cite-paper	5,343
	author	5,959	1,902	paper-ref-paper	5,343
	subject	56	1,902	paper-author	9,949
	term	1,902	—	paper-subject	3,025
				paper-term	255,619
Freebase				(+3 inverse relations)	
	book *	40,402	film 19,427	music 82,351	
	sports	1,025	people 17,641	location 9,368	
	organization	2,731	business 7,153	no features at all	
	36 typed relations	(BOOK-AND-BOOK 202,674, MUSIC-AND-MUSIC 283,670, ...)			

* labelled target type. Labelled counts: DBLP 4,057/4,057 authors; ACM 3,025/3,025 papers; IMDB 4,573/4,932 movies (of which **2,684 carry more than one label**); Freebase 7,954/40,402 books. Split is 24% train / 6% validation / 70% test on the labelled target nodes.

Provenance of Table 5. The HGB paper publishes only the totals of Table 4; it never prints a per-type breakdown. The numbers above were therefore recomputed directly from the released `node.dat/link.dat/label.dat` files, and every one of the four datasets reproduces its published totals exactly (nodes, edges, number of node types, number of edge types and number of classes all match). That agreement is what licenses the table. Two incidental findings: HGB’s own `info.dat` declares the DBLP author attribute dimension as 0, but the data file actually carries 334-dimensional author features — and 334 is exactly the feature dimension HAN publishes for DBLP, which independently corroborates the data over the metadata; and ACM’s 1,902-dimensional bag-of-words is indexed by exactly the 1,902 `term` nodes.

2.2 The individual benchmark graphs

DBLP (bibliographic). A computer-science bibliography extract with types *author*, *paper*, *term*, *venue/conference*. Authors are labelled with one of 4 research areas (database, data mining, AI, information retrieval); author features are a bag-of-words over the keywords of the papers they wrote. It is the canonical author-classification HIN and the standard testbed for APA vs. APCPA semantics.

ACM (bibliographic). Papers from ACM venues with types *paper*, *author*, *subject*, *term*. Papers are labelled with 3 classes (database, wireless communication, data mining) and carry a bag-of-words feature. Distinctively, ACM has a *paper–paper* citation relation in addition to the star structure, and HGB splits it into a citation and a reference direction — so the schema is not a pure star.

IMDB (movie). Types *movie*, *director*, *actor* and (in HGB) *keyword*. Movies are labelled by genre. This is the benchmark where versions diverge most sharply: MAGNN’s IMDB has 3 *disjoint* genres (Action/Comedy/Drama), whereas HGB’s has 5 *overlapping* genres, making it multi-label.

Freebase. *The name denotes two unrelated graphs.* HGB’s Freebase is a general knowledge graph: 8 node types, 36 relations, 180,098 nodes, target type BOOK with 7 classes, and **no node features whatsoever** — so it isolates how much a model can extract from pure typed structure. It is *book-centric* only in the sense that BOOK is the labelled type; its schema is not a star but is in fact *complete* (Figure 5). HeCo’s “Freebase” is instead a *movie* dataset (3,492 movies, 33,401 actors, 2,502 directors, 4,459 writers; target movie, 3 classes). Quoting a Freebase number without saying which one is meaningless.

AMiner (bibliographic). Two scales. metapath2vec’s AMiner is web-scale: 1,693,531 authors, 3,194,405 papers, 3,883 venues, with 8 Google-Scholar categories as labels and 246,678 labelled authors. HeCo’s AMiner is a small subset (6,564 papers, 13,329 authors, 35,890 references; target paper, 4 classes). *Erratum worth knowing:* metapath2vec’s body text says the dataset “consists of 9,323,739 computer scientists”, but the authors’ own correction note states that 9,323,739 is the paper–author *link* count and the author count is 1,693,531. The wrong figure is still widely re-quoted.

Yelp and Douban (business / review). HERec’s Yelp has types *user* (16,239), *business* (14,284), *city* (47), *category* (511) and *compliment* (11), with U–B 198,397, U–U 158,590, U–Co 76,875, B–Ci 14,267 and B–Ca 40,009 edges (density 0.08%). Douban Movie has *user* (13,367), *movie* (12,677), *director* (2,449), *actor* (6,311) and *type* (38). Both are used for rating prediction / recommendation, reported with MAE and RMSE rather than F1.

Amazon (e-commerce) — heterogeneous in $\mathcal{R}$ only. HGB’s Amazon is the limiting case of the definition in Section 1, and worth stating for that reason. It has **one** node type (*product*, 10,099 nodes) and **two** relation types, *co-viewing* and *co-purchasing*, over 148,659 edges — the electronics subset preprocessed by GATNE (Cen et al., KDD 2019). So $|\mathcal{A}| = 1$ but $|\mathcal{R}| = 2$, giving $|\mathcal{A}| + |\mathcal{R}| = 3 > 2$: it is a heterogeneous network even though every node is the same type, which is precisely why the condition in Section 1 is stated as a sum rather than as “more than one node type”. Its network schema is therefore a single node carrying two distinct self-relations. It is used for **link prediction only** (it has no node labels), scored with ROC-AUC and MRR. It is also the cleanest illustration of the limit of meta-path methods: with one node type there is no type alternation to define a meta-path over, and HGB’s link-prediction table accordingly leaves MAGNN’s Amazon entry blank, noting that vacancies are “due to lack of meta-paths on those datasets”. Relation-aware but meta-path-free models (R-GCN, GATNE, simple-HGN) are the ones that can run on it at all.

LastFM and MovieLens (user–item HINs). MAGNN’s Last.fm has *user* (1,892), *artist* (17,632) and *tag* (1,088), with U–U 12,717, U–A 92,834 and A–T 23,253; it carries *no labels and no features*, so it is a pure link-prediction benchmark (predict user–artist), with one-hot IDs as dummy inputs. MovieLens/ML-100K is used analogously as a user–item bipartite HIN enriched with occupation/genre types. **PubMed**, in its HGB heterogeneous form, is a 4-type biomedical graph (63,109 nodes, 10 relation types) used for disease–disease link prediction — not to be confused with the homogeneous PubMed citation network of Problem 1.

OGB: ogbn-mag and MAG240M. These supply the scale the older benchmarks lack. **ogbn-mag** is a Microsoft Academic Graph subset with *paper* (736,389), *author* (1,134,649), *institution* (8,740) and *field of study* (59,965) — 1,939,743 nodes — and four relations: *author–affiliated_with–institution* (1,043,998), *author–writes–paper* (7,145,660), *paper–cites–paper* (5,416,271) and *paper–has_topic–field_of_study* (7,505,078), totalling 21,111,007 edges. The task is predicting each paper’s **venue** among **349 classes**; only *paper* nodes have features (128-d word2vec), the rest are featureless. Crucially the split is **temporal** — train ≤ 2017 , validation 2018, test ≥ 2019 — not random, so it measures generalisation forward in time. The official metric is accuracy. **MAG240M** (OGB-LSC) is the same design two orders of magnitude larger: 121,751,666 papers, 122,383,112 authors and 25,721 institutions (244,160,499 nodes) with 1,728,364,232 edges, 768-d RoBERTa features, and 153 arXiv subject classes.

2.3 Version incompatibility — the caveat that matters

The same dataset name denotes different graphs in different papers. Table 6 makes this concrete; quoting a statistic without naming the version is an error.

Four specific discrepancies are worth flagging.

Table 6: The same dataset name denotes different graphs in different papers. Figures that differ from the majority version are bolded.

Dataset	Version	Node counts	Note
DBLP	HAN	paper 14,328 / 14,327, author 4,057, term 8,789 , venue 20	P-T = 88,420
	MAGNN / HeCo / HGB	paper 14,328, author 4,057, term 7,723 , venue 20	P-T = 85,810
	GTN	18,405 total (= paper + author + venue; <i>term dropped</i>)	4 edge types
ACM	HAN	paper 3,025, author 5,835, subject 56	P-A = 9,744
	HGB	paper 3,025, author 5,959 , subject 56, term 1,902	P-A = 9,949
	HeCo	paper 4,019 , author 7,167 , subject 60	P-A = 13,407
IMDB	HAN	movie 4,780, director 2,269, actor 5,841	
	MAGNN	movie 4,278, director 2,081, actor 5,257	3 classes, single-label
	HGB	movie 4,932, director 2,393, actor 6,124, key-word 7,971	5 classes, multi-label

1. **DBLP’s term count.** HAN reports 8,789 terms; MAGNN, HeCo and HGB all use 7,723. HGB’s published total is only consistent with the latter ($4,057 + 14,328 + 7,723 + 20 = 26,128$, whereas the HAN figure would give 27,194). HAN’s own table is also internally inconsistent: it lists 14,328 papers in the Paper–Author and Paper–Conference rows but 14,327 in the Paper–Term row.
2. **HGB’s ACM does not match its stated provenance.** HGB says it uses HAN’s subset while “preserving all edges including paper citations and references”, but its author count (5,959 vs. 5,835) and P–A edge count (9,949 vs. 9,744) differ from HAN’s published table. Re-adding citation edges cannot change the number of *authors*, so the two ACM graphs genuinely differ.
3. **IMDB changes task type, not just size.** HGB’s IMDB is multi-label with 5 overlapping genres; MAGNN’s is single-label with 3 disjoint ones. An F1 on one is not comparable to an F1 on the other even in principle (Section 5).
4. **HGB’s LastFM edge count double-counts one relation.** Its node total 20,612 exactly equals MAGNN’s $1,892 + 17,632 + 1,088$, but its edge total is 141,521 against MAGNN’s 128,804. The gap is exactly 12,717, the U–U count: $2 \times 12,717 + 92,834 + 23,253 = 141,521$. So the symmetric user–user relation is stored in both directions while U–A and A–T are stored in one.

3. Network schemas and meta-paths

Each schema in Figures 4 and 5 is $T_G = (\mathcal{A}, \mathcal{R})$ for one benchmark family: circles are node types (letter = the initial used in meta-path notation), edges are relation types. Inverse relations are omitted for readability in Figure 4; Freebase (Figure 5) has no inverse relations at all and is drawn complete.

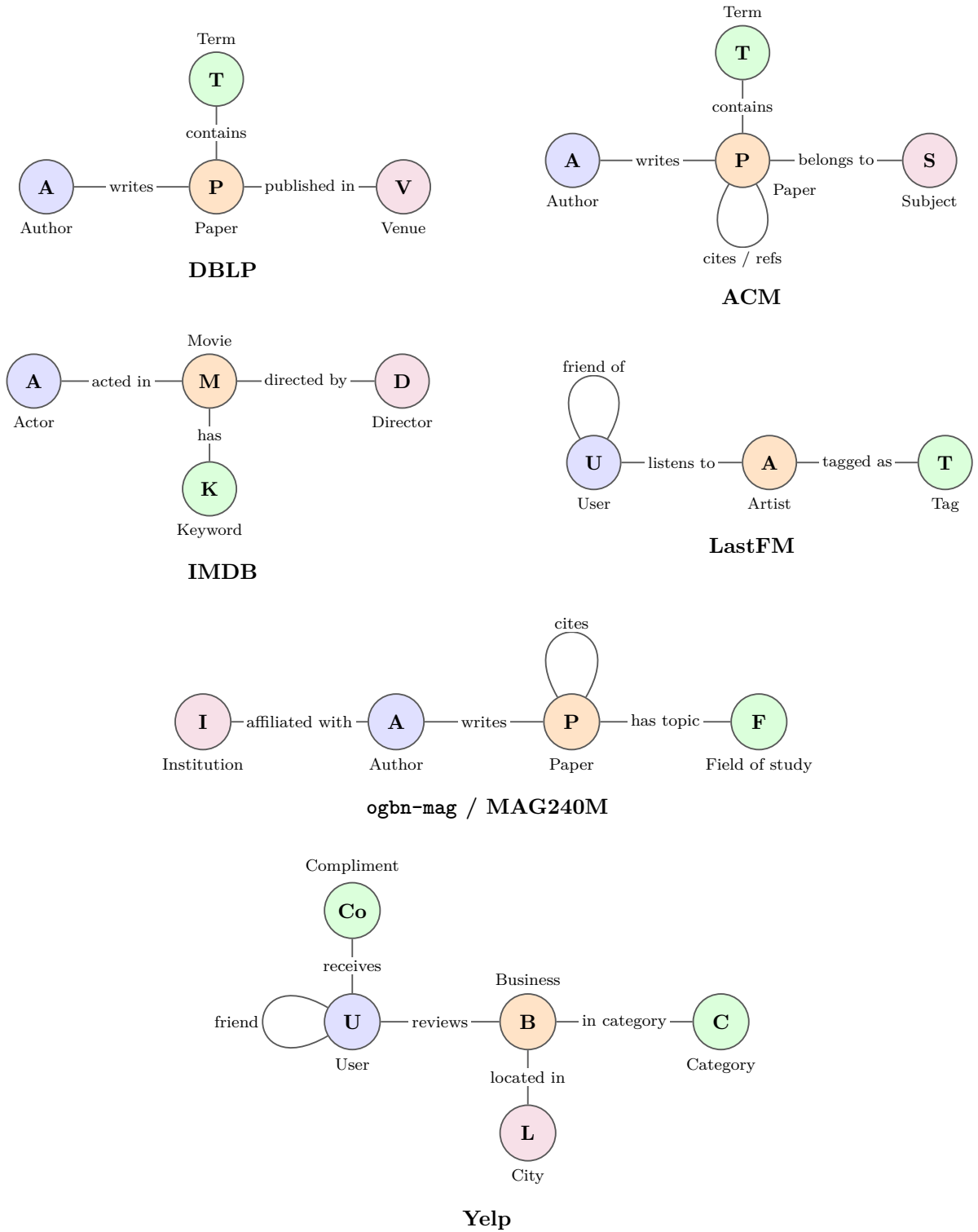


Figure 4: Network schemas $T_G = (\mathcal{A}, \mathcal{R})$ of the bibliographic, media, large-scale and review benchmarks. Circles are node types (the letter is the initial used in meta-path notation); labelled edges are relation types. Inverse relations are omitted in this figure — but see Figure 5, where nothing is omitted.

Table 7: Meta-paths commonly used on each benchmark, and the semantic relation each encodes.

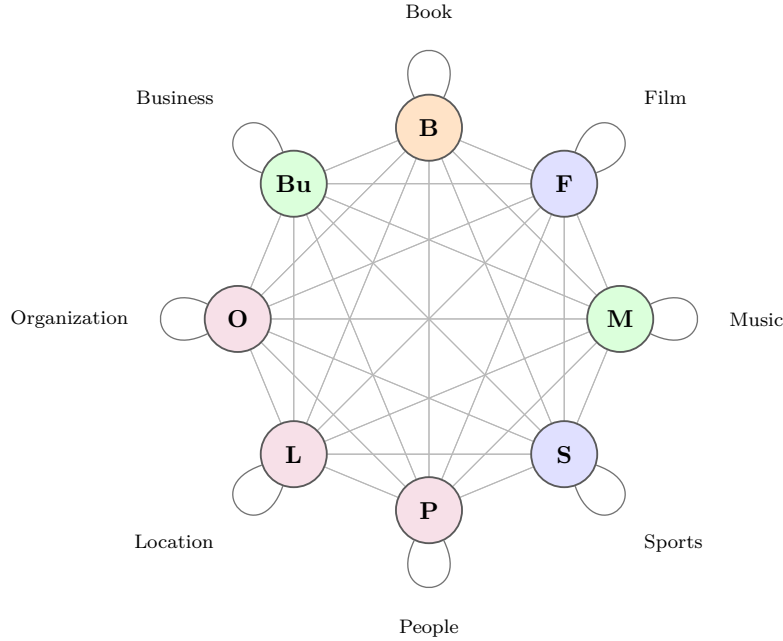
Dataset	Meta-paths	Semantics
DBLP	APA	two authors co-wrote a paper (co-authorship)
	APCPA / APVPA	two authors published at the same venue (community)
	APTPA	two authors used the same term (topical similarity)
ACM	PAP	two papers share an author
	PSP	two papers share a subject
	PcPAP, PrPSP	HGB variants routed through a citation (c) or reference (r) hop
IMDB	MAM	two movies share an actor
	MDM	two movies share a director
	AMA, DMD	two actors co-starred / two directors share a movie
ogbn-mag	PAP	two papers share an author
	PAIAP	two papers from the same institution (via author-institution)
LastFM	UAU	two users listen to the same artist
	UATAU	two users like artists sharing a tag
	AUA, ATA	two artists share a listener / a tag
Yelp	UBU	two users reviewed the same business
	UBCaBU	two users reviewed businesses of the same category
Freebase	BB, BFB, BPB	book-book; books sharing a film / a person

4. Baselines

4.1 Homogeneous methods, reported as type-agnostic baselines

These are reported in essentially every heterogeneous paper, and the *point* of reporting them is that they **discard ϕ and ψ entirely** — they run on the graph with all type information erased (or on one meta-path-induced projection). They are the control condition: whatever a heterogeneous model gains over them is the measured value of modelling types. HGB’s central and uncomfortable finding is that this gap is much smaller than the literature had claimed, and that a properly tuned GAT beats most specialised HGNNs.

- **DeepWalk** — Perozzi, Al-Rfou & Skiena, KDD 2014. Truncated uniform random walks treated as sentences, fed to skip-gram with hierarchical softmax.
- **LINE** — Tang, Qu, Wang, Zhang, Yan & Mei, WWW 2015. Explicitly preserves first-order (observed edge) and second-order (shared-neighbourhood) proximity, optimised with negative sampling and edge-sampling for large graphs.
- **node2vec** — Grover & Leskovec, KDD 2016. DeepWalk with a biased second-order walk whose p, q parameters interpolate between BFS-like (structural equivalence) and DFS-like (homophily) exploration.
- **GCN** — Kipf & Welling, ICLR 2017. First-order spectral convolution: a layer is a symmetrically normalised neighbourhood average followed by a linear map and nonlinearity.



Freebase (HGB) — the schema is *complete*: $\binom{8}{2} = 28$ pairwise relations + one self-relation per type = 36.

Figure 5: Network schema of Freebase (HGB), drawn *in full* — all 36 relation types are shown, so unlike Figure 4 nothing is omitted here. Freebase is the one benchmark whose schema is **complete**: every one of the $\binom{8}{2} = 28$ unordered type pairs carries exactly one relation, and every type carries one self-relation, giving $28 + 8 = 36$. Note the contrast with DBLP/ACM/IMDB: **none of these 36 are inverses** — all 36 are distinct forward relations, so this count is *not* inflated by the forward/inverse doubling of Section 1, whereas DBLP’s “6 edge types” is 3 relations plus their 3 inverses. BOOK (shaded) is the labelled target type; the largest single relation is MUSIC–MUSIC, 283,670 edges or 27% of the graph.

- **GAT** — Veličković, Cucurull, Casanova, Romero, Liò & Bengio, ICLR 2018. Replaces the fixed normalisation with multi-head self-attention over neighbours. (Commonly miscited as 2017, the arXiv posting date.)
- **GraphSAGE** — Hamilton, Ying & Leskovec, NIPS 2017. Inductive framework that samples a fixed number of neighbours and aggregates with mean/LSTM/max-pool, enabling embeddings for unseen nodes.

4.2 Meta-path and random-walk HIN embedding

- **metapath2vec** — Dong, Chawla & Swami, KDD 2017. Random walks are *meta-path-guided*: at each step the walker must move to the node type prescribed by the meta-path scheme (e.g. APVPA), so the generated context is type-balanced instead of being dominated by the most frequent type. The walks feed a heterogeneous skip-gram. Note that the base model still normalises negative sampling over *all* nodes regardless of type; it is the **metapath2vec++** variant that normalises the softmax per node type.
- **HIN2Vec** — Fu, Lee & Lei, CIKM 2017. Rather than fixing meta-paths in advance, it learns

node *and* relation (meta-path) vectors jointly by training a multi-task binary classifier that predicts, for a node pair, which relations hold between them.

- **ESim** — Shang, Qu, Liu, Kaplan, Han & Peng, arXiv:1610.09769, 2016 (*preprint only — never published in a peer-reviewed venue*). Embeds nodes by sampling positive and negative path instances of user-supplied weighted meta-paths and optimising a noise-contrastive objective for similarity search.
- **HERec** — Shi, Hu, Zhao & Yu, IEEE TKDE 31(2):357–370, **2019**. Performs meta-path-guided random walks, but first *filters* each walk to the target type, producing homogeneous sequences per meta-path; the resulting embeddings are fused by a nonlinear function and injected into an extended matrix-factorisation recommender.
- **PTE** — Tang, Qu & Mei, KDD 2015. Decomposes a heterogeneous text network (word–word, word–document, word–label) into bipartite graphs and jointly embeds them with the LINE objective, giving semi-supervised, task-tuned text embeddings.

4.3 Heterogeneous graph neural networks

- **R-GCN** — Schlichtkrull, Kipf, Bloem, van den Berg, Titov & Welling, ESWC **2018**. Relation-specific message passing: each layer aggregates neighbours separately per relation r with its own weight matrix W_r , normalised by $c_{i,r} = |\mathcal{N}_i^r|$, plus a self-loop term. Because $|\mathcal{R}|$ weight matrices overfit, it offers two *alternative* regularisers — *basis decomposition* (W_r a linear combination of B shared bases) and *block-diagonal decomposition* (W_r a direct sum of small dense blocks). For link prediction it is used as an encoder with a **DistMult** decoder.
- **HAN** — Wang, Ji, Shi, Wang, Ye, Cui & Yu, WWW 2019. Two-level attention: *node-level* GAT-style attention within each meta-path-induced neighbour set, then *semantic-level* attention learning a scalar importance per meta-path to fuse them. Its limitation — the one MAGNN attacks — is that it operates on meta-path-induced graphs, so intermediate nodes along the meta-path are discarded.
- **MAGNN** — Fu, Zhang, Meng & King, WWW 2020. Three stages: node-content transformation projecting differently-dimensioned type attributes into a shared space; *intra*-meta-path aggregation that encodes the **entire path instance including intermediate nodes** (mean, linear, or a relational-rotation encoder borrowed from RotatE) with attention over instances; and *inter*-meta-path semantic attention.
- **GTN** — Yun, Jeong, Kim, Kang & Kim, NeurIPS 2019. *Learns* meta-paths instead of accepting them. A 1×1 convolution with softmax weights forms a soft convex combination Q_i of the per-edge-type adjacency matrices; multiplying $A = Q_1 Q_2 \cdots Q_l$ yields new meta-path adjacencies differentially. Including the identity among the candidates lets it learn paths *shorter* than l . A GCN then runs on the learned graph.
- **HetGNN** — Zhang, Song, Huang, Swami & Chawla, KDD 2019. Restart-based random walks sample a fixed-size, type-balanced neighbour set; content of each node is encoded with a Bi-LSTM over its heterogeneous attributes, neighbours are aggregated per type, and types are combined by attention.
- **HGT** — Hu, Dong, Wang & Sun, WWW 2020. Node-type-specific projections produce Key/Query/Value, and attention and message matrices are indexed by the *meta-relation*

$\langle \tau(s), \phi(e), \tau(t) \rangle$, with a learnable scalar prior μ per meta-relation. It needs no hand-designed meta-paths. Adds *relative temporal encoding* for dynamic graphs and *HGSampling* for web-scale mini-batch training.

- **simple-HGN / HGB** — Lv, Ding, Liu et al., KDD 2021. A deliberately minimal baseline: GAT plus (i) a learnable **edge-type embedding** concatenated into the attention logit, (ii) residual connections on *both* node representations and attention scores, and (iii) L_2 normalisation of the output embeddings. The paper’s larger contribution is the HGB benchmark itself and the finding that this simple model matches or beats most specialised HGNNs.
- **ie-HGCN** — Yang, Guan, Li, Zhao, Cui & Wang, IEEE TKDE 35(2):1637–1650, 2023. Hierarchical object-level then type-level aggregation which, as a by-product, *ranks* the useful meta-paths, making the model interpretable while avoiding an exhaustive meta-path search.
- **RSHN** — Zhu, Zhou, Pan, Zhu & Wang, IEEE ICDM 2019. Builds a coarsened *edge-centric* “relation structure graph” to embed relation types themselves, then propagates with those relation embeddings rather than treating relations as bare labels.
- **HPN** — Ji, Wang, Shi, Wang & Yu, IEEE TKDE 35(1):521–532, 2023. Analyses semantic confusion (the HIN analogue of over-smoothing) and replaces the propagation step with a personalised-PageRank-like update that retains a fraction of the node’s own representation, enabling deeper meta-path propagation.

4.4 Self-supervised / contrastive heterogeneous methods

- **DMGI** — Park, Kim, Han & Yu, AAAI 2020. Extends Deep Graph Infomax to *attributed multiplex* networks: one DGI mutual-information objective per relation type, with a consensus regularisation that pulls the per-relation embeddings towards a shared representation.
- **HDGI** — Ren, Liu, Huang, Dai, Bo & Zhang, arXiv:1911.08538, 2019 (AAAI-20 DLGMA workshop, under a different title; *no archival venue*). Applies mutual-information maximisation between local node embeddings and a global summary, using meta-path-based convolution plus semantic attention as the encoder.
- **HeCo** — Wang, Liu, Han & Shi, KDD 2021. Contrasts two *views* of the same node: a *network-schema* view (attention over directly-connected neighbours of each other type) and a *meta-path* view (meta-path-specific GCNs with semantic attention). Positives are not merely the node’s counterpart view — HeCo selects multiple positives by counting how many meta-path instances connect two nodes and taking the top ones.

4.5 Knowledge-graph embeddings as relation-aware LP baselines

A HIN with many relation types is close to a knowledge graph, so heterogeneous link-prediction papers (and HGB’s LP track) routinely report KG embedding models, which score a typed triple (h, r, t) directly.

- **TransE** — Bordes, Usunier, Garcia-Duran, Weston & Yakhnenko, NIPS 2013. Models a relation as a translation, $\mathbf{h} + \mathbf{r} \approx \mathbf{t}$, scoring $-\|\mathbf{h} + \mathbf{r} - \mathbf{t}\|$. Cannot represent 1-to-N or symmetric relations.

- **DistMult** — Yang, Yih, He, Gao & Deng, ICLR 2015. Bilinear score $\langle \mathbf{h}, \text{diag}(\mathbf{r}), \mathbf{t} \rangle$; efficient but necessarily *symmetric*.
- **ComplEx** — Trouillon, Welbl, Riedel, Gaussier & Bouchard, ICML 2016. Moves DistMult into $\mathbb{C}$ and takes the real part of a Hermitian product, so asymmetric relations become representable at the same cost.
- **ConvE** — Dettmers, Minervini, Stenetorp & Riedel, AAAI 2018. Reshapes h and r into 2-D and applies a convolution, giving a parameter-efficient, highly expressive scorer.
- **RotatE** — Sun, Deng, Nie & Tang, ICLR 2019. Treats a relation as an element-wise *rotation* in complex space, $\mathbf{t} = \mathbf{h} \circ \mathbf{r}$ with $|r_i| = 1$, capturing symmetry, antisymmetry, inversion and composition simultaneously.

5. Evaluation metrics

Two conventions govern the whole literature. First, a HIN benchmark normally labels **exactly one node type** — the author in DBLP, the paper in ACM, the movie in IMDB — so “accuracy on DBLP” always means accuracy on that one target type, the other types being context. Second, for *unsupervised* methods the protocol is **embed** $\rightarrow$ **freeze** $\rightarrow$ **fit a linear classifier**: the embedding is learned without labels, then held fixed while a logistic regression is trained on a fraction of target nodes, the fraction varied over 20/40/60/80% and mean $\pm$ std reported over repeated splits. HAN, MAGNN and HeCo all report in this form.

5.1 Node classification

With class set $\mathcal{C}$ and per-class counts $\text{TP}_c, \text{FP}_c, \text{FN}_c$, **Micro-F1** pools the confusion matrix *before* scoring,

$$\begin{aligned} \text{Micro-P} &= \frac{\sum_c \text{TP}_c}{\sum_c (\text{TP}_c + \text{FP}_c)}, & \text{Micro-R} &= \frac{\sum_c \text{TP}_c}{\sum_c (\text{TP}_c + \text{FN}_c)}, \\ \text{Micro-F1} &= \frac{2 \text{Micro-P} \cdot \text{Micro-R}}{\text{Micro-P} + \text{Micro-R}}, \end{aligned}$$

whereas **Macro-F1** scores each class and averages *unweighted*:

$$\text{F1}_c = \frac{2 \text{TP}_c}{2 \text{TP}_c + \text{FP}_c + \text{FN}_c}, \quad \text{Macro-F1} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \text{F1}_c.$$

In the *single-label* multi-class setting every misclassification creates one false positive and one false negative, so $\text{Micro-P} = \text{Micro-R}$ and Micro-F1 equals plain **accuracy**. The pair is always reported together because they answer different questions: Micro-F1 is dominated by frequent classes, Macro-F1 weights a rare class as heavily as a common one, and a large gap between them signals a model that has learned the class prior rather than the graph.

The multi-label caveat. This equivalence **fails** for HGB’s IMDB and Freebase tracks, where a node may carry several labels at once. Prediction is per-label thresholding (sigmoid + binary cross-entropy) rather than an arg max, and the pooled confusion matrix is taken over the flattened node $\times$ label indicator matrix; there $\text{Micro-P} \neq \text{Micro-R}$ in general and Micro-F1 is *not* accuracy. In HGB’s IMDB, 2,684 of the 4,573 labelled movies carry more than one genre, so this is not a corner case — and it is why a Micro-F1 on HGB’s IMDB cannot be compared with one on MAGNN’s single-label IMDB.

Other classification metrics. **Accuracy** is the official metric on OGB (ogbn-mag, MAG240M). **ROC-AUC** is used for binary or strongly imbalanced targets (and by HeCo alongside F1). For the *clustering* probe that HAN, MAGNN, DMGI and HeCo all run — k -means on the frozen embeddings with $k = |\mathcal{C}|$ — the scores are

$$\text{NMI}(Y, \hat{C}) = \frac{2I(Y; \hat{C})}{H(Y) + H(\hat{C})} \in [0, 1], \quad \text{ARI} = \frac{\text{RI} - \mathbb{E}[\text{RI}]}{\max(\text{RI}) - \mathbb{E}[\text{RI}]},$$

with I mutual information, H entropy and RI the Rand index. Both are 0 in expectation for a chance partition and 1 for a perfect one (ARI may go negative). These test whether the embedding geometry is class-structured with no classifier fitted at all.

5.2 Link prediction

The essential difference from the homogeneous case is that in a HIN link prediction is stated **per relation type**: not “is there an edge” but “is there an edge of type R ” — predict Author–Paper, or Paper–Venue, or User–Artist. Two consequences follow. (i) *Negatives must be sampled type-consistently*: a negative for an Author–Paper edge must join an author to a paper, never an author to a venue, since a type-violating negative is trivially rejected by any type-aware model and inflates its apparent margin over type-agnostic baselines. HGB samples negatives from 2-hop neighbours at a 1:1 ratio for exactly this reason. (ii) Scores are reported *per relation* and then averaged, so a model can be strong on a dense relation and useless on a sparse one without that showing in a single number.

Let rank_i be the position of the i -th true edge among its candidates, over query set Q :

$$\text{MRR} = \frac{1}{|Q|} \sum_{i \in Q} \frac{1}{\text{rank}_i}, \quad \text{Hits@K} = \frac{1}{|Q|} \sum_{i \in Q} \mathbf{1}[\text{rank}_i \leq K],$$

i.e. MRR is the mean reciprocal rank of the first true item and Hits@ K the fraction of positive edges ranked inside the top K against their sampled negatives ($K \in \{1, 3, 10\}$ conventionally, inherited from the KG literature). **ROC-AUC** equals the probability that a random positive outranks a random negative. **Average precision** (equivalently AUPRC) is

$$\text{AP} = \sum_n (R_n - R_{n-1}) P_n,$$

with P_n, R_n precision and recall at the n -th threshold. AP is preferred to ROC-AUC when positives are rare — always true for link prediction on a sparse graph — because ROC-AUC is optimistic there, the enormous true-negative count sitting in its denominator. **Precision@K / Recall@K**, and MAE/RMSE for rating prediction, appear in the recommendation-flavoured HIN papers (HERec on Yelp/Douban). HGB’s LP track standardises on **ROC-AUC and MRR**.

5.3 Summary: dataset → target → task → metric

Closing remark. The single most important methodological point is the one HGB was created to make: because DBLP, ACM and IMDB circulate in mutually incompatible preprocessed versions (Table 6), and because IMDB even changes from single-label to multi-label between versions, numbers copied across papers are frequently not comparable. Any credible evaluation must name the dataset *version*, the split, and whether the protocol is end-to-end supervised or embed-freeze-classify.

Table 8: Dataset → target node type → task → metric normally reported.

Dataset	Domain	Target	Task	Labels	Metrics normally reported
DBLP	bibliographic	author	NC	4, single	Macro-F1, Micro-F1 (+NMI/ARI)
ACM	bibliographic	paper	NC	3, single	Macro-F1, Micro-F1 (+NMI/ARI)
IMDB (HGB)	movie	movie	NC	5, multi	Macro-F1, Micro-F1
IMDB (MAGNN)	movie	movie	NC	3, single	Macro-F1, Micro-F1
Freebase (HGB)	KG	book	NC	7, single	Macro-F1, Micro-F1
AMiner	bibliographic	author/paper	NC	8 / 4	Macro-F1, Micro-F1, NMI
ogbn-mag	bibliographic	paper	NC	349, single	Accuracy (official)
MAG240M	bibliographic	paper	NC	153, single	Accuracy (official)
LastFM	music	user–artist	LP	—	ROC-AUC, MRR
Amazon	e-commerce	product–product	LP	—	ROC-AUC, MRR
PubMed (HGB)	biomedical	disease–disease	LP	—	ROC-AUC, MRR
Yelp / Douban	review	user–item	LP / rating	—	Prec@K, Rec@K; MAE, RMSE
OAG (HGT)	bibliographic	paper–field/venue	LP	—	NDCG, MRR